Puffin UAV Pro User Manual

Edition: 5-1-2026

Overview of Puffin Product

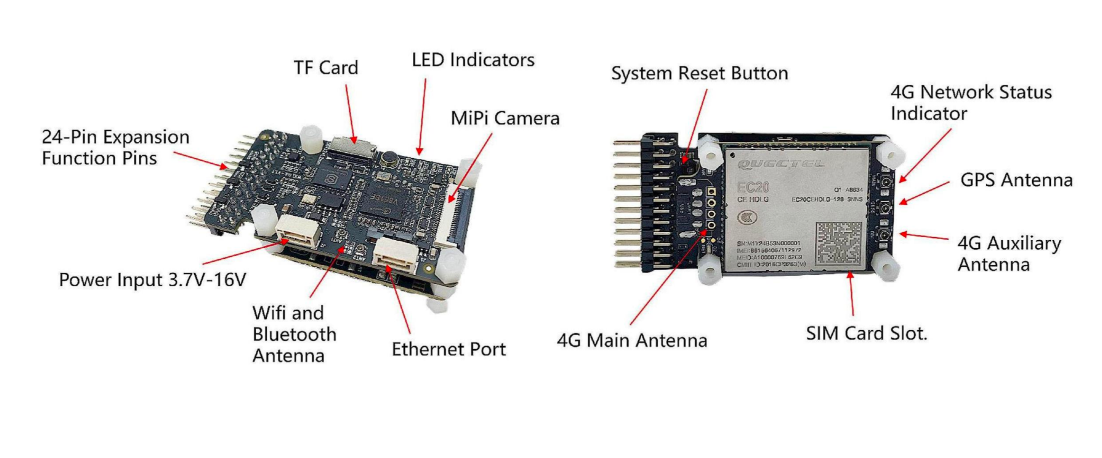
Puffin is a lightweight remote control module for unmanned devices (video transmission, data, control). Relying on the 4G cellular network and adopting WebRTC P2P technology, it achieves ultra-low latency video transmission (~100ms RTT) and control (~30ms RTT).

CPU	**Cortex-A7 core@900MHz** **RISC-V@600MHz**
NPU/ DDR	On-chip 0.5T NPU On-chip 64M DDR
Operating System	Tina-Linux
4G LTE Cellular Network Connection	Cellular Network: 4G LTE MIMO, Upstream: Below 50Mbps Line: 150Mbps
Other wireless connections	WIFI: 2.4G Bluetooth: 2.4G GNSS: GPS/GLONASS

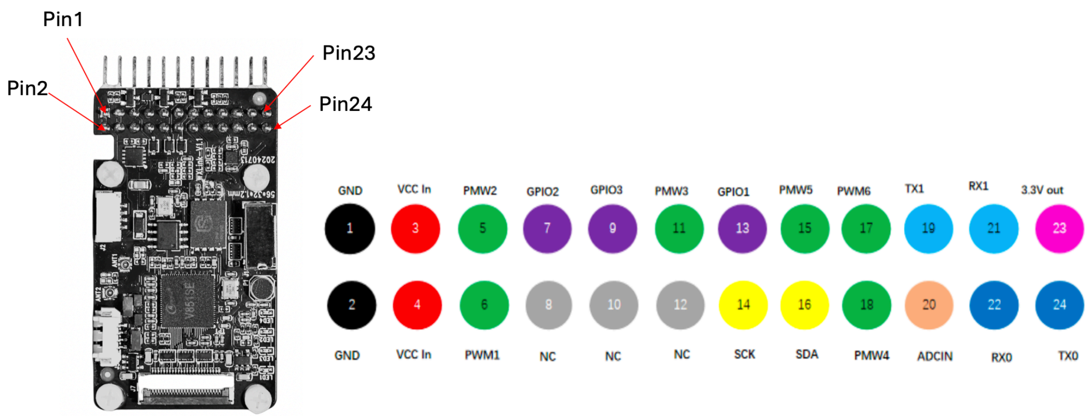
Puffin hardware

Hardware Interface Description

1. Hardware Interface Description



2. Serial connection with the drone flight controller

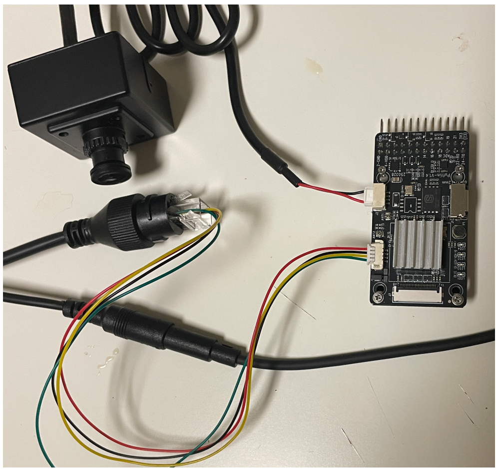


First, you need to familiarize yourself with the 24 PIN pin expansion interface of TEL801. When connecting to the flight controller, you only need to connect GND, TX1 (pin19), and RX1 (21) to the corresponding ports of the flight controller's UART (TELEM). Note that if the flight controller's UART port can output a stable voltage (4-16V), you can connect Vcc to directly power Puffin through the flight controller's UART port. Otherwise, please do not connect the Vcc lead, and Puffin needs to be powered from another power source.

The pins most relevant to our current configuration are as follows:

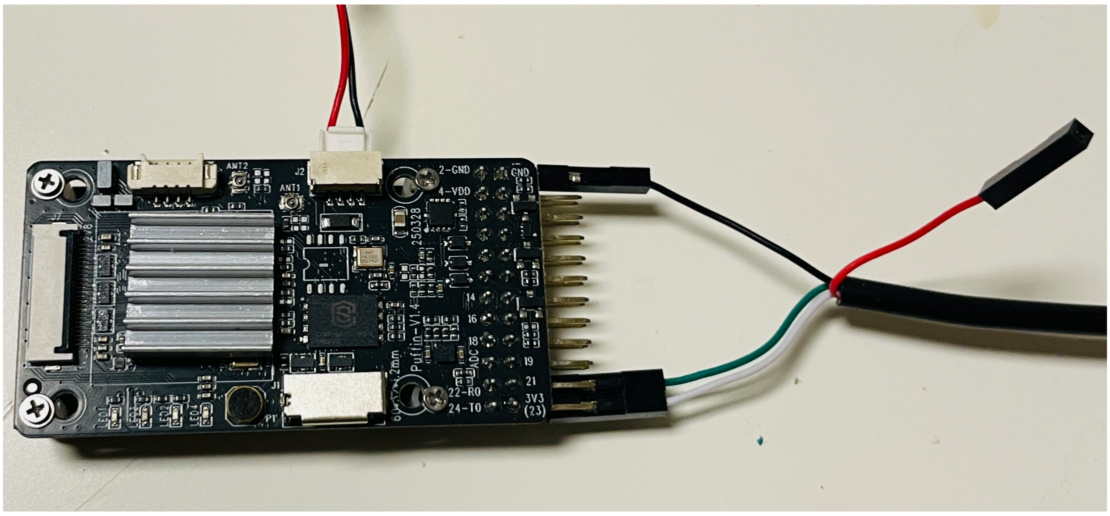
- GND (Ground): Pin 1, Pin 2
- Vcc in (Power Input): Pin 3, Pin 4
- Serial Port (2 sets):
  - Pins 22 and 24 Debug Serial Port
  - Pins 19 and 21 are connected to the serial port of the flight controller

3. Connect RTSP Camera (Gimbal)



The network port of Puffin can be connected to the RTSP protocol camera of a drone. The hardware connection method is as follows: Connect the 4-pin network port of Puffinto the RTSP camera with a standard network port using the included 4-pin network port adapter cable, as shown in the figure. The RTSP camera requires separate power supply.

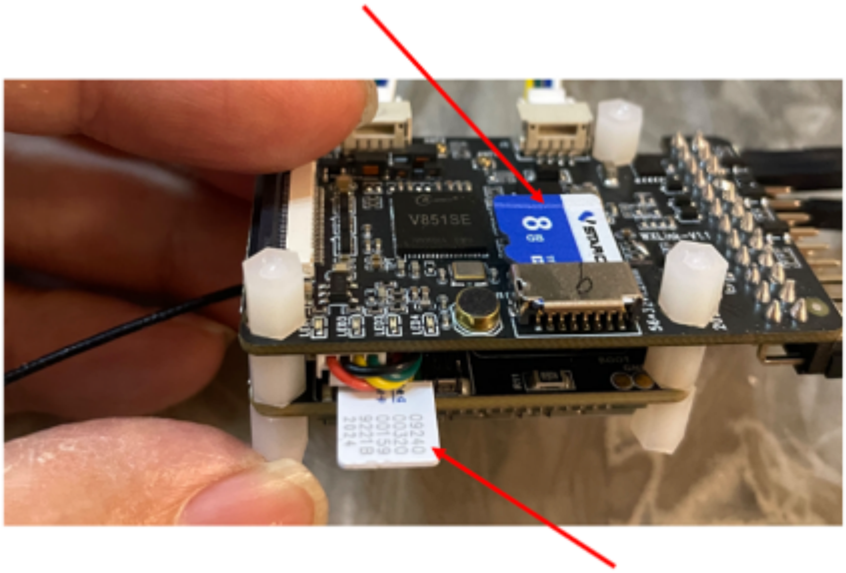
When the Puffinmodule connects to an RTSP camera, the default IP Address is : rtsp://192.168.0.153:554/ch01.264. If users need to connect to an RTSP camera with their own address, they need to modify the configuration file inside the Puffinfirmware, It is very important to set the network segment of the RTSP camera to segment 0. The method is as follows:



4. SD Card and SIM Card Insertion

First, insert the SD card into the designated card slot. This SD card will be used to store important data and files required for operation. If using the 4G version, please also insert the SIM card into the corresponding card slot. It is recommended to use your own daily mobile data SIM card, do not use IoT SIM cards! (They may have data limits) Be sure to note that you must insert and remove the SD card and SIM card while the device is powered off! Otherwise, it may cause permanent damage to the hardware!

MicroSD card

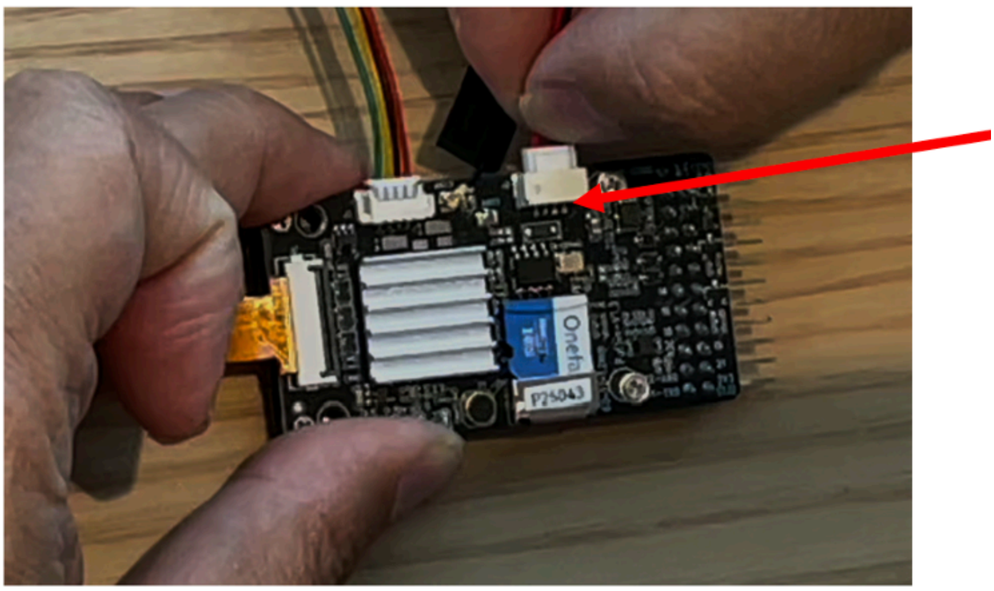


Sim Card be aware of Orientation

5. Power Supply

Puffinsupports multiple power supply methods and 4-16V DC voltage.

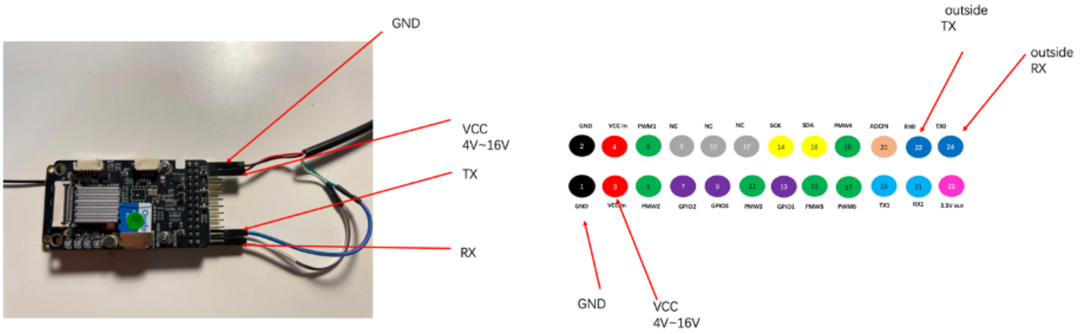
5.1 Interface Power Supply



As shown in the figure above, connect to a power outlet using the provided USB cable. It is recommended to use an independent 5V power supply.

5.2 Connect to external lithium battery power supply

The external lithium battery can be directly connected to VCC (pins 3/4) and GND (pins 1/2) of the 24-pin expansion pins, with an input voltage range of 4-16V.



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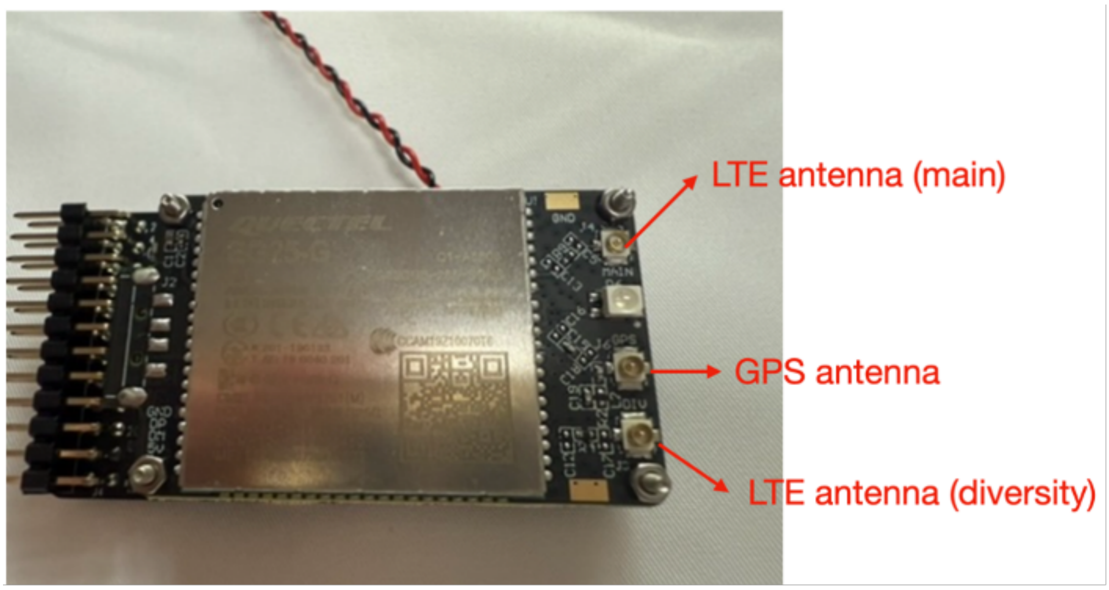
6. Antenna Installation

6.1 LTE/GPS Antenna

The LTE antenna and GPS antenna are located on the lower layer of the laminated board. LTE uses dual MIMO antennas.



6.2 Antenna Fixed Position

Be sure to pay attention to the installation position of the antenna. The antenna we provide is an FPC antenna. Do not directly attach the antenna to metal or carbon fiber sheets, as the antenna performance will be greatly affected! The antenna can be fixed on plastic components or suspended. Note that many fuselages are made of carbon fiber, which has a shielding effect, **do not directly fix the antenna to the carbon fiber fuselage!** A plastic foam at least 5 cm thick can be used to isolate the antenna from the fuselage.

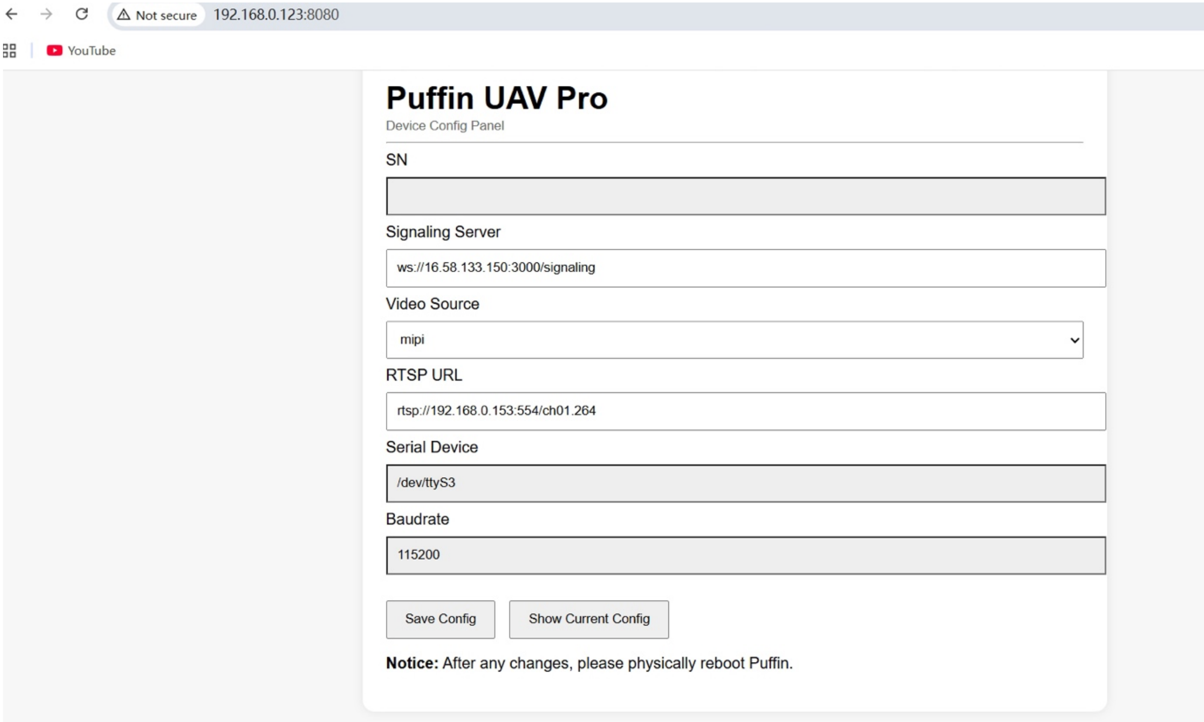

TE 4G Full_Band

Puffin

Ground Control

Software Package

1. Configuration Change



Network and Video Configuration

You may need to configure the **signaling server address** and the **RTSP camera IP address** on the Puffin board.

To access the configuration interface:

1. Connect the Puffin board to your PC using the provided Ethernet cable.
2. Open a web browser and enter:

   ```
   192.168.0.123:8080
   ```

3. The configuration interface will appear.

If you are using your own private signaling server, make sure to update the signaling server address accordingly.

Video Source Selection

Puffin supports two types of video input sources:

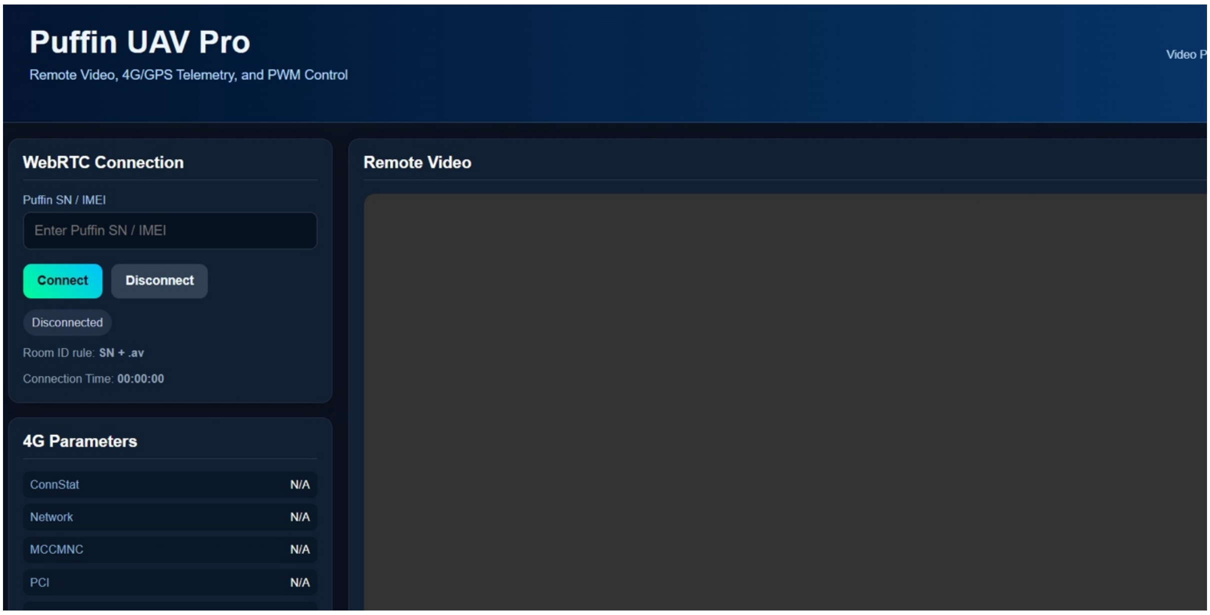
- **MIPI camera** (included)
- **RTSP camera** (external)

If you choose to use an RTSP camera:

- Ensure that the RTSP URL configured on the Puffin board matches the actual camera stream URL.
- It is strongly recommended to verify the RTSP stream using a player such as VLC media player before testing with Puffin.
- If the stream works correctly in VLC, it should also work with Puffin.

Note: Puffin only supports **H.264 encoded video streams**.

2. Ground Control – Video& PWM Interface



Puffin_UAV-5-1-26 supports:

- **4-channel PWM output control**
- **Video input via MIPI or RTSP camera**

To access the ground control interface:

1. Navigate to the `gc_video` folder.
2. Double-click the HTML file to open it in your browser.
3. Enter your device **SN (Serial Number)**.
4. Click **Connect**.

Once connected, you will be able to:

- View **real-time video streaming**
- Monitor **4G network parameters**

The most important network metric is **RSRP (Reference Signal Received Power)**, which indicates cellular signal strength.

If a GPS antenna is connected, the interface will also display **real-time GPS coordinates**.

PWM Control

The interface provides control for **4 PWM output channels** via on-screen sliders.

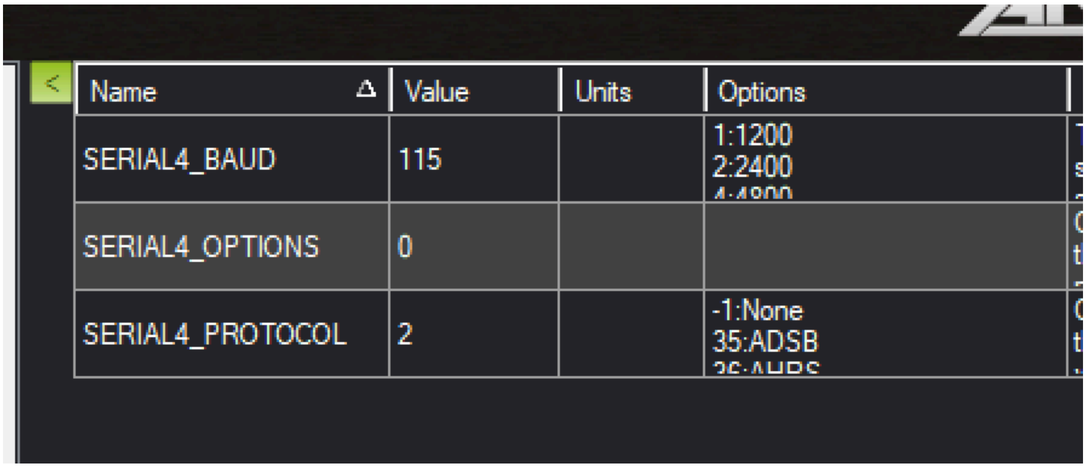
Customization

If you require a customized UI or additional features, please contact our sales team.

3. Puffin Connects to Flight Controller

3.1 Flight Control Parameter Settings

First, set the correct flight controller parameters. Connect the flight controller to Mission Planner. Find the corresponding UART port in config->Full parameter list. If using ArduPilot and Serial Port 4 is used, set the parameters in Serial4 as shown in the figure below.



Name	Value	Units	Options
SERIAL4_BAUD	115		1:1200 2:2400
SERIAL4_OPTIONS	0		
SERIAL4_PROTOCOL	2		-1:None 35:ADSB

Baud rate:115200

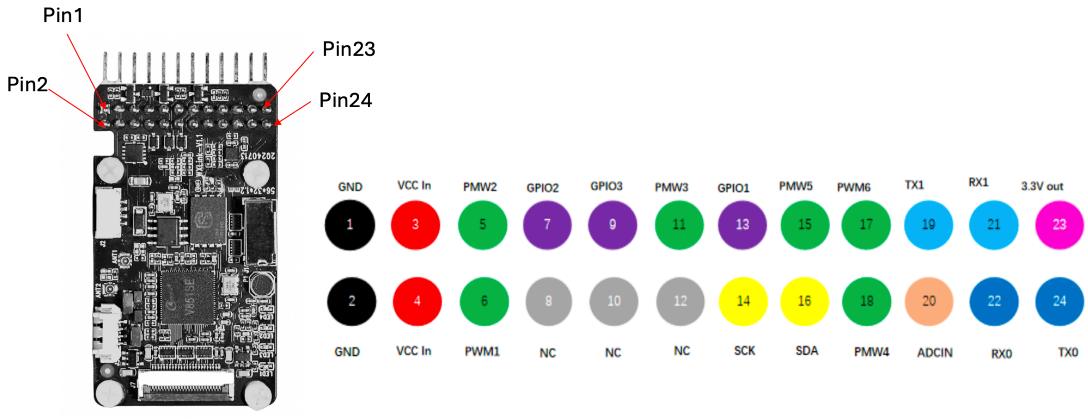
Options:0

Protocol:2

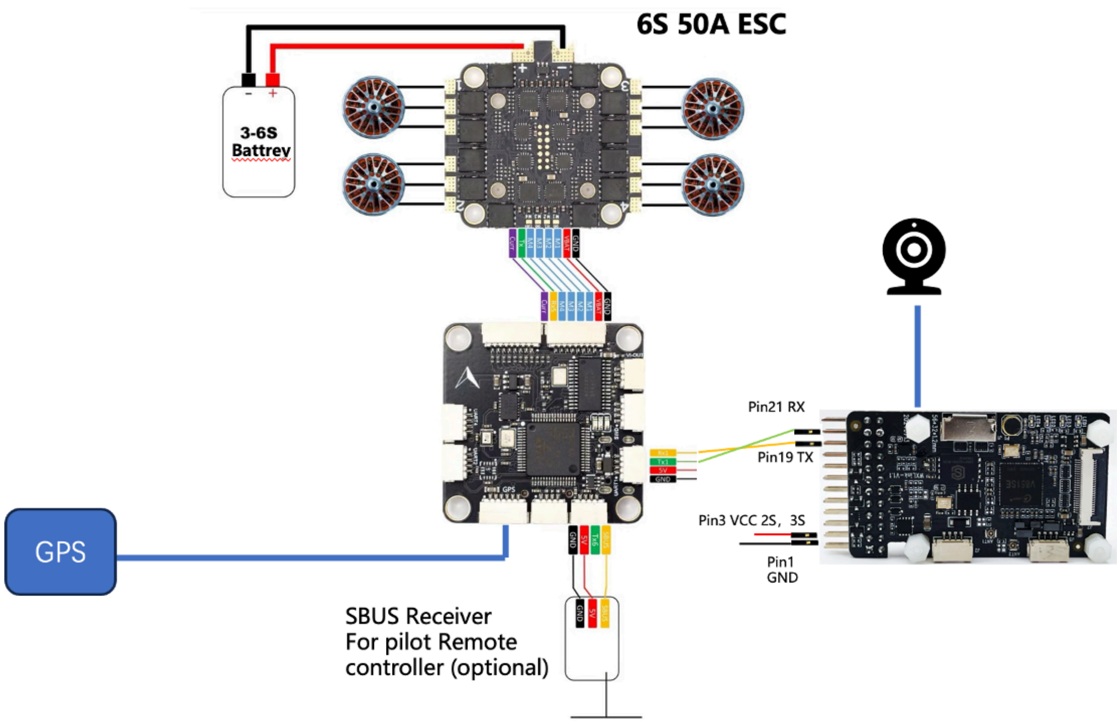
After the settings are completed, click "write params", then power off and power on the flight controller to restart it, and confirm that the parameters are set to the correct values. The same principle applies to PX4 flight controllers. Confirm that the UART port of the flight controller supports the Mavlink2 protocol.

3.2 Connect the flight controller to Puffin

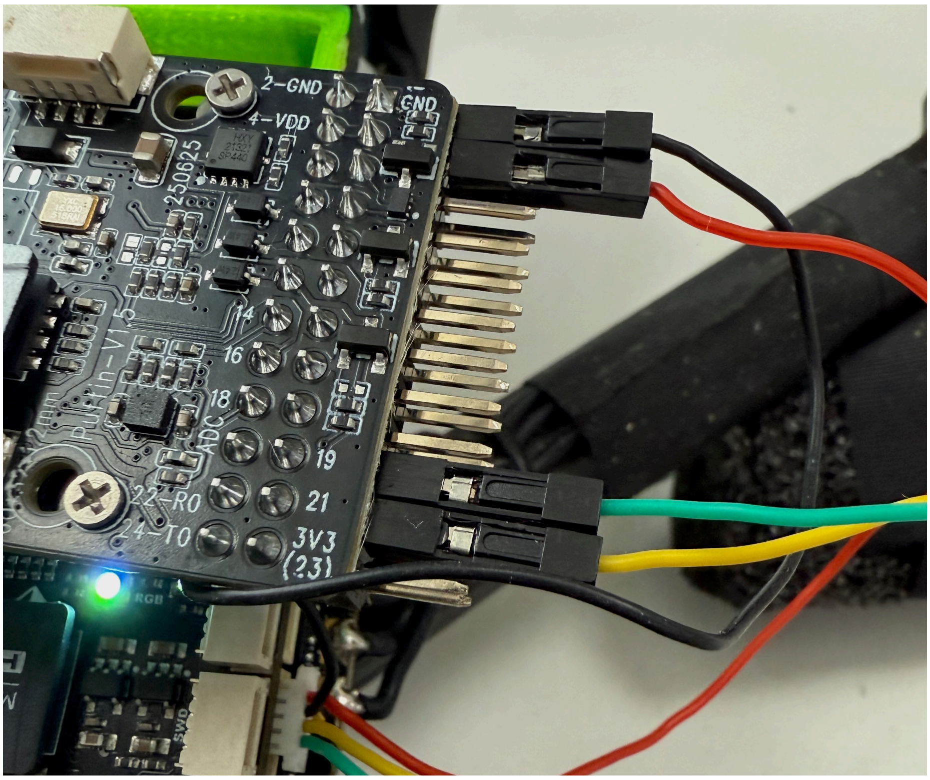
The flight controller is connected to Puffin via serial port, with the principle being serial port pass-through. As shown in the figure below, the TX line of the flight controller UART needs to be connected to the RX (pin21) of Puffin, and the RX of the flight controller is connected to the TX (pin19) of Puffin. It is necessary to connect the GND of the flight controller UART to the GND of Puffin. Whether to connect Vcc depends on the power supply situation of Puffin Board. If the flight controller UART serial port can output a stable voltage, it can be connected and used as the power supply for Puffin. If Puffin already has an external power supply, then it MUST NOT connect Vcc.



Flight Controller: Existing models owned by the user, supporting serial communication

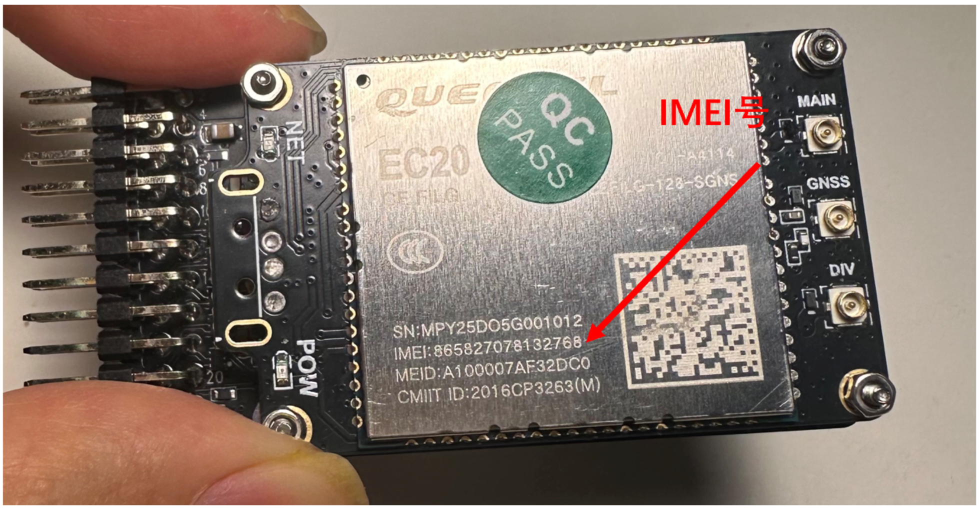


The following figure shows the connection between Puffin board and MircoAir743-V2 flight controller. The serial port of MicroAir743 supports a constant voltage of 5V, which can be used as the power supply for Puffin.



3.3 Get device IMEI number

The IMEI number of this device can be seen below the metal casing of the 4G module on the back of the module



3.4 Camera Connection:

Puffin supports RTSP network cameras, requiring cameras that can change their IP addresses. The network port of Puffin is connected to the network camera. Please set the IP address of the network camera to rtsp://192.168.0.153:554/ch01.264, with the encoding format being H.264, resolution 1080P, and frame rate 30fps. If the camera cannot change its IP settings, please contact customer after-sales service.

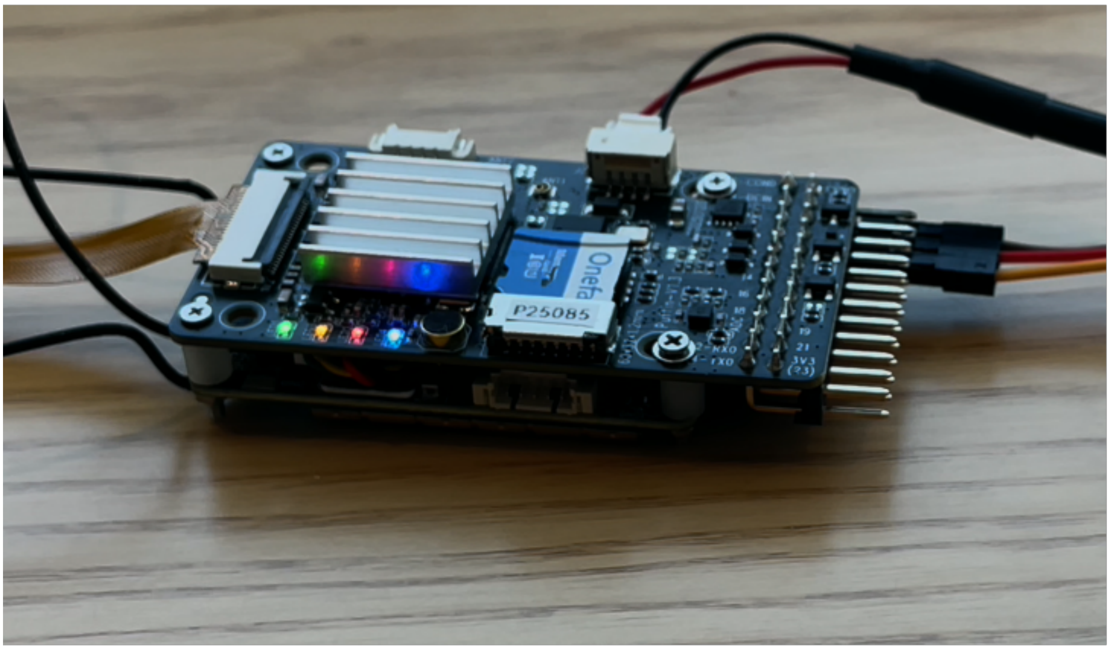
3.5 Check List

Please confirm before powering on

- SIM card inserted correctly, MicroSD card inserted correctly
- 4G Antenna Installed Correctly
- Serial port is correctly connected to the flight controller
- The network port is correctly connected to the pod (if the pod requires external power supply, pay attention to powering it on).

3.6 Power On

Power on Puffin. Puffin can be powered by the 2S to 4S battery of the drone, with the maximum power supply voltage not exceeding 18V. Pay attention to observing the blue LED light in the figure below. A blinking blue light indicates successful network connection. Otherwise, please power off and then power on again.



After the power cord is connected to the power supply for a dozen seconds, ensure that the red LED on the front of the module remains constantly lit, and the blue LED flashes, indicating that the module has successfully connected to the signaling server. If the blue LED does not flash, please check if the SIM card has data. If it does not flash after power-on, please contact technical support.

Blue Light Frequently Asked Questions

- The SIM card has run out of data
- Antenna not installed
- The 4G signal in the indoor environment is very weak

4. Puffin connects the flight controller and the ground station

Download the software package

Currently, the ground station only supports the Windows version of QGroundControl and Mission Planner. On a Windows PC with the ground station installed. The folder mavlink_connect is for connection with Mission Planner and QGC

Ground Station Access (QGC, MP)

1. Enter **mavlink_connect** folder, edit **conf.yaml** file

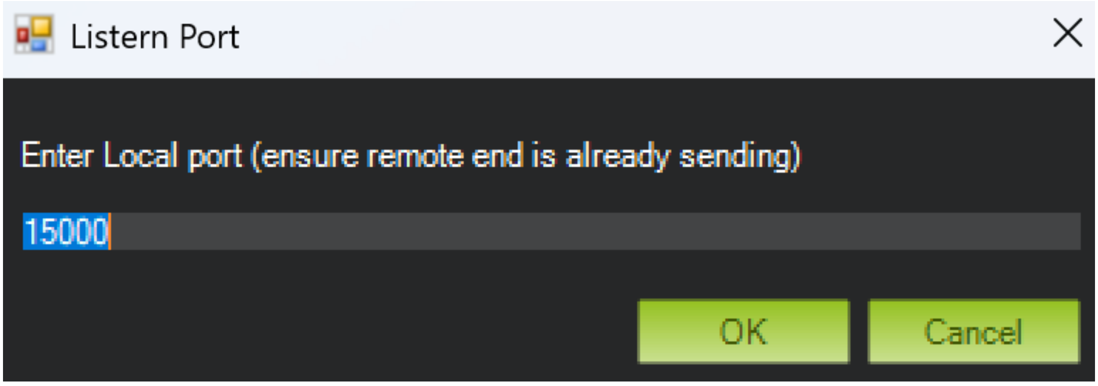
2. Edit the number after "sn:" to the IMEI number of the 4G module's metal shell on the back of the Puffinmodule. For example, sn: 867652073902932.

3. For international version make sure the server IP address is rtcServer: 16.58.133.150 in the file, as this is the IP Address of the test signaling server provided by Puffin.

4. Start **UAVGCS_RTCReconnect.cmd** Program

5. On the PC interface, the CMD program running window and running log will appear. If you see "Connected" within the first 10 lines of the log, it indicates that the serial port signal of the flight controller has been pass-through transmitted to this PC, and the Puffinmodule has established a control data connection channel.

6. Open Mission Planner, select UDP 115200 connection, and choose local port 15000,

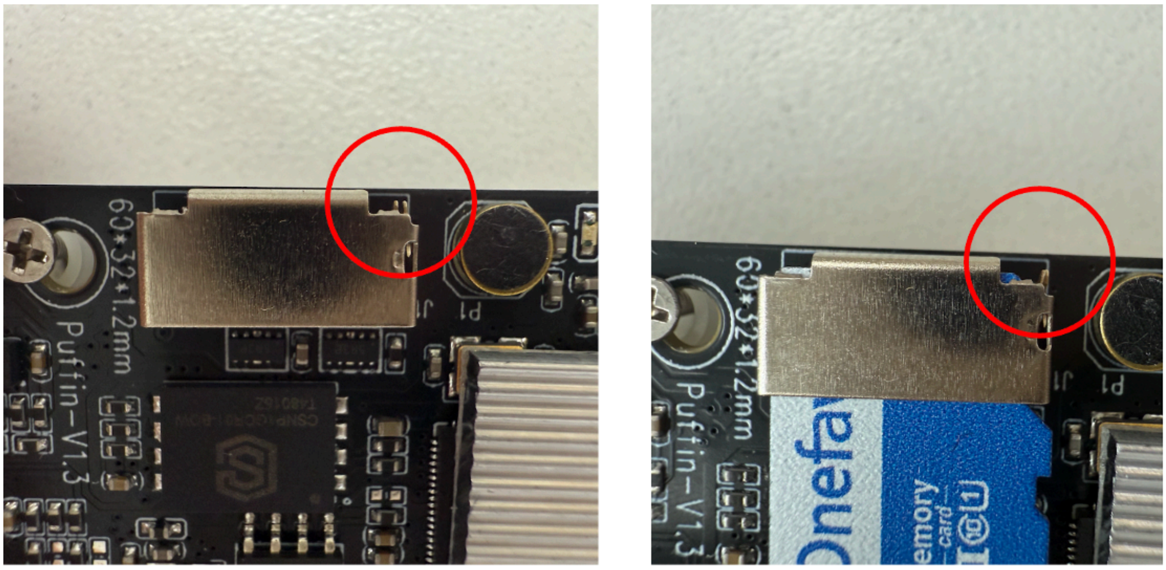


7. After a few seconds, it can be seen that the ground station can display the telemetry information of the flight control system in real time.

Debug

1. SD card is not inserted securely

Many unsuccessful cases are due to the SD card not being fully inserted into the card socket, causing the entire system to fail to boot up. Observe the reed on the upper right corner of the SD card socket. If the card is fully inserted, the reed will be in a closed state. If you confirm that the card has been inserted but the reed is not closed, you can use tweezers to pinch it.



No card state: Reed open V.S. Card present state: Reed closed (short circuit)